



TRAIN PROBLEMS?

Basic To Advance

- by YASH JAIN



TRAIN PROBLEMS?



Why Study This Topic?



- CAT
- XAT
- CMAT
- SNAP
- NMAT
- MAT
- IIFT
- GMAT
- GATE
- ESE
- Placements
- Government Exams
- Civil Services Exams
- Banking Exams
- Railway Exams
- College Entrance Exams
- Air Force Exams (AFCAT)
- SSC
- MHCET
- GRE
- Speed Time & Distance has 2 segments, this segment is easy & scoring.
- Short & Important Topic, hence scoring topic
- In Railways Exams & AMCAT oriented exams, you can expect more questions from here



BACK TO
BASICS

UNIT CONVERSIONS (kmph to mps)

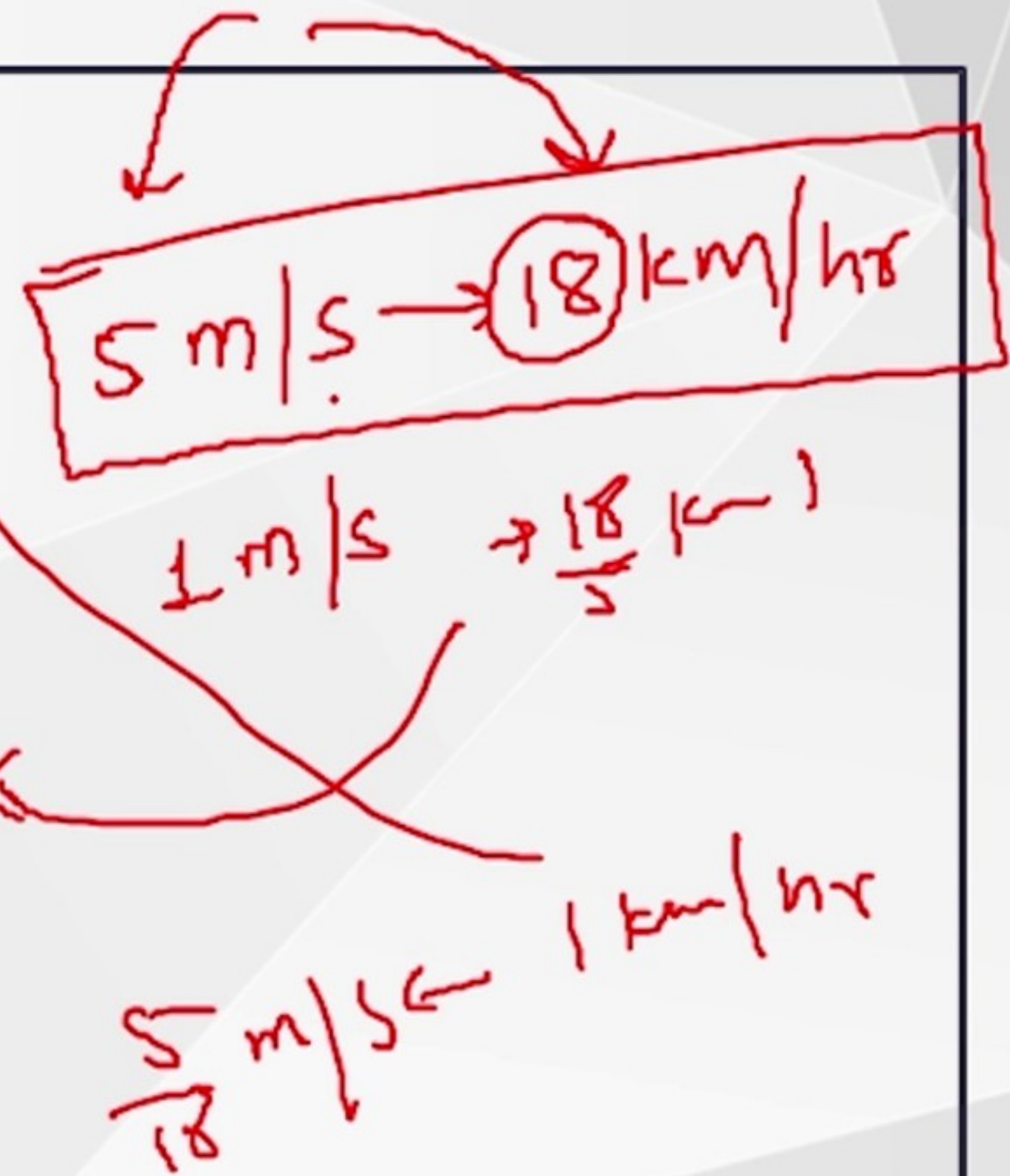
$$1 \text{ km/hr} = 5/18 \text{ m/s}$$

$$1 \text{ m/s} = \text{18/5} \text{ km/hr}$$

UNIT CONVERSIONS (kmph to mps)

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/hr}$$



UNIT CONVERSIONS (kmph to mps)

$$5 \text{ m/s} \rightarrow 18 \text{ km/hr}$$

18 km/hr

$\times 2$

5 m/s

$$18 \times \frac{5}{18} = 5 \text{ m/s}$$

36 km/hr

$\times 3$

10 m/s

54 km/hr

15 m/s

54 km/hr

54 km/hr

15 m/s

kmph
mps

$\times 3$
18 km/hr

5 m/s

UNIT CONVERSIONS (kmph to mps)

63 km/hr

72 km/hr

45 km/hr

$$\cancel{63} \times \frac{5}{18}$$

$$20 \text{ m/s}$$

$$36 + 9$$

$$10 \text{ m/s} + 2.5 \text{ m/s}$$

$$\underline{12.5 \text{ m/s}} \checkmark$$

$$\textcircled{45} \text{ km/hr}$$

$$63 \Rightarrow 54 + 9$$

$$15 \text{ m/s} + 2.5 \text{ m/s}$$

$$17.5 \text{ m/s}$$

$$\frac{5 \text{ m/s} \rightarrow 18 \text{ km/hr}}{\times 3 \quad \downarrow \times 3}$$



Case 1:

Train Crossing a Pole/Signal/Stationary objects



Que: Punjab Mail Express of length 80 meters goes to Delhi to visit Baba Ka Dhaba.
Find the time taken (in seconds) by Punjab Mail Express to cross a pole if it is travelling with a speed of 80 km/hr?



This is copyrighted content of KNOWLEDGE GATE EDUVENTURES, piracy of this video & sharing course contents & credentials will lead to filing of police case & heavy fine.

Que: Punjab Mail Express of length 80 meters goes to Delhi to visit Baba Ka Dhaba.
Find the time taken (in seconds) by Punjab Mail Express to cross a pole if it is travelling with a speed of 80 km/hr?

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$80 \Rightarrow \frac{5}{18} \times 80$$

$$S = \frac{d}{t}$$

$$t = \frac{d}{S}$$

$$t = \frac{\text{length of train}}{\text{Speed}}$$

$$= \frac{80 \text{ metres}}{\frac{5}{18} \times 80}$$

$$= \frac{18}{5} = \frac{36}{10} = 3.6 \text{ sec.}$$

Que: Punjab Mail Express of length 80 meters goes to Delhi to visit Baba Ka Dhaba. Find the time taken (in seconds) by Punjab Mail Express to cross a pole if it is travelling with a speed of 80 km/hr?

$$\text{Length (m)} = \text{Speed (km/hr)}$$

Pole/signal

3.6 seconds

Que: Find the time taken (in seconds) by Bundelkhand Express (length 100 meters) to cross a pole if it is travelling with a speed of 100 km/hr ?



This is copyrighted content of KNOWLEDGE GATE EDUVENTURES, piracy of this video & sharing course contents & credentials will lead to filing of police case & heavy fine.

Que: Find the time taken (in seconds) by Bundelkhand Express (length 100 meters) to cross a pole if it is travelling with a speed of 100 km/hr ?

3.6 seconds



This is copyrighted content of KNOWLEDGE GATE EDUVENTURES, piracy of this video & sharing course contents & credentials will lead to filing of police case & heavy fine.

Que: Find the time taken (in seconds) by Bundelkhand Express (length 100 meters) to cross a pole if it is travelling with a speed of 100 km/hr ?

$$s = \frac{d}{t}$$
$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$100 \text{ km/hr} = \frac{5}{18} \times 100 \text{ m/s}$$

$$t = \frac{d}{s}$$

$$= \frac{100 \text{ m}}{\frac{5}{18} \times 100 \text{ m/s}}$$

$$= \frac{18}{5} \text{ sec} = \underline{\underline{3.6 \text{ seconds}}}$$



Que: Grand Trunk Express (120 meters long) crosses an electric pole in 8 seconds. Find the speed of Grand Trunk Express in kmph ?



This is copyrighted content of KNOWLEDGE GATE EDUVENTURES, piracy of this video & sharing course contents & credentials will lead to filing of police case & heavy fine.

Que: Grand Trunk Express (120 meters long) crosses an electric pole in 8 seconds. Find the speed of Grand Trunk Express in kmph? 54 kmph

$$S = \frac{d}{t}$$
$$= \frac{120}{8} \text{ m/s}$$
$$= 15 \text{ m/s}$$
$$15 \text{ m/s} \rightarrow 18 \text{ km/hr}$$
$$\downarrow \times 3$$
$$54 \text{ km/hr}$$



This is copyrighted content of KNOWLEDGE GATE EDUVENTURES, piracy of this video & sharing course contents & credentials will lead to filing of police case & heavy fine.

Que: Punjab Mail Express of length 80 meters goes to Delhi to visit Baba Ka Dhaba. Find the time taken (in seconds) by Punjab Mail Express to cross a pole if it is travelling with a speed of 80 km/hr?

$$S = \frac{\text{distance}}{\text{time}}$$

$$\text{relative speed} = \frac{\text{sum}}{\text{time to meet}}$$

Principal : ye teacher khaali period me tumhe padhane aaye hai to tum inhe pareshan kyu kar rhe ho ?

Whole class :

